

Partial differential equations

$$\text{If } z = f(x, y)$$

$$\frac{\partial z}{\partial x} = p$$

$$\frac{\partial z}{\partial y} = q$$

$$\frac{\partial^2 z}{\partial x^2} = r$$

$$\frac{\partial^2 z}{\partial x \partial y} = s$$

$$\frac{\partial^2 z}{\partial y^2} = t$$

Partial differential eqⁿ

Let $f(x, y, z, x^2 + y^2 + z^2) = 0$ (three variables)

$$Pp + Qq = R$$

$$P(1, 2)$$

$$Q(3, 1)$$

$$R(1, 2)$$

$$\begin{matrix} x & y & z \\ 1 & 2 & 3 \end{matrix}$$

Lagrange's linear PDE

Equation of the type

$$[Pp + Qq = R] \quad (1)$$

is called Lagrange's linear PDE

The subsidiary or auxiliary equations are

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

The solution of eqⁿ (1) is given by

$$f(u, v) = 0$$

where u & v are fⁿ of

x, y & z .

Euler theorem.

If $f(x, y)$ is a homogeneous function of x & y in degree n ,

then according to Euler theorem

$$x \frac{df}{dx} + y \frac{df}{dy} = n f$$

Jacobian

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

Maxima & Minima

⇒ Working process: If $f(x, y)$ be a f^n of 2 variable x & y then

i) find $\frac{\partial f}{\partial x}$ & $\frac{\partial f}{\partial y}$

ii) the necessary condition for maxima & minima

$\frac{\partial f}{\partial x} = 0$ & $\frac{\partial f}{\partial y} = 0$, find the

pts a, b

iii) find:

$$r = \frac{\partial^2 f}{\partial x^2}, \quad s = \frac{\partial^2 f}{\partial x \partial y}, \quad t = \frac{\partial^2 f}{\partial y^2}$$

at the point a, b & calculate the value of $r^2 - s^2$

iv) If $x^2 - y^2 > 0, x > 0$ then $f(x, y)$ has the minimum at the point (a, b)

v) If $x^2 - y^2 > 0, x < 0$ then $f(x, y)$ has a maxima at the point (a, b)

vi) If $x^2 - y^2 < 0$ then $f(x, y)$ has neither maxima nor minima at point (a, b)

vii) If $x^2 - y^2 = 0$ then test fail further investigation req

Lagrange's method of undetermined multipliers

Acc. to Lagrange's method

$$f(x, y, z) = \underline{x y^2 z^3} + \lambda (x + y + z - 24)$$

↓
vii) khamama
ya minima
karna hai
is se

↓
vii
given
hai